

Finite-time and finite-size scalings of coercivity in dynamic hysteresis

Miao Chen^{1,2}, Xiu-Hua Zhao^{1,2,*} and Yu-Han Ma^{1,2,3,†}

¹*School of Physics and Astronomy, Beijing Normal University, Beijing 100875, China*

²*Key Laboratory of Multiscale Spin Physics (Ministry of Education), Beijing Normal University, Beijing 100875, China*

³*Graduate School of China Academy of Engineering Physics, Beijing 100193, China*



(Received 7 August 2025; accepted 20 February 2026; published 19 March 2026)

The coercivity landscape for characterizing hysteresis in interacting systems across multiple timescales is proposed by Chen *et al.* in a companion paper [Phys. Rev. Lett. **136**, 117102 (2026)]. For the stochastic ϕ^4 model under periodic driving of rate v_H , the coercivity landscape $H_c(v_H)$ exhibits plateau features at a characteristic rate v_p with the corresponding coercivity H_p . Below this plateau ($v_H < v_p$), the $H_c \sim v_H$ scaling obtained in the near-equilibrium regime becomes inaccessible in the thermodynamic limit. Above the plateau ($v_H > v_p$), scaling in the fast-driving regime, $H_c \sim v_H^{1/2}$, is completely different from that, $H_c - H_p \sim (v_H - v_p)^{2/3}$, in the postplateau slow-driving regime. The emergence of the plateau with a finite-size scaling reflects the competition between the thermodynamic limit and the quasistatic limit. In this paper, we provide detailed analytical proofs and numerical evidence supporting these results. Moreover, to demonstrate the coercivity landscape in concrete physical systems, we study the magnetic hysteresis in the Curie-Weiss model and analyze its finite-size effects. We reveal that finite-time coercivity scaling shows model-specific behavior only in the fast-driving regime, while exhibiting universal characteristics elsewhere.

DOI: [10.1103/mzvj-n6vn](https://doi.org/10.1103/mzvj-n6vn)

I. INTRODUCTION

Hysteresis refers to the lag in a system's response relative to its equilibrium state when subjected to an external driving field. The history dependence of hysteresis is closely associated with irreversibility and ergodicity breaking [1,2]. In the quasistatic limit, where the external field changes infinitely slowly, one expects the system to follow the equilibrium curve without hysteresis. However, this expectation breaks down for systems undergoing a first-order phase transition (FOPT). For mean-field FOPT models with long-range interactions, the system in the thermodynamic limit is governed by deterministic dynamics and therefore fails to transition from the metastable to the stable phase, becoming trapped in a metastable state characterized by local equilibrium [3–5]. Even when FOPT proceeds via stochastic nucleation and growth, the system may still exhibit dynamical ergodicity breaking, owing to the weakness of thermal fluctuations relative to the nucleation barrier [6,7].

Numerous studies have explored the scaling behavior of hysteresis with respect to the driving rate of the external field, both when quasistatic hysteresis is present and absent. In most cases, the area enclosed by the hysteresis loop serves as a quantitative measure of hysteresis [3,8–12]. In systems without quasistatic hysteresis, a linear scaling of the hysteresis area is a common result in the near-equilibrium regime [13]. In contrast, for systems undergoing FOPT with quasistatic hysteresis, it has been shown that the difference in hysteresis

area between the finite-time case and quasistatic limit scales as $2/3$ -power-law scaling in the slow-driving regime [3,14]. Interestingly, both types of scaling have been observed within the same model under different conditions. Wu *et al.* [15] studied the scaling relation of extra work in the Curie-Weiss model and revealed scaling crossover that reflects the interplay between finite-time dynamics and finite-size effects. Moreover, numerical and experimental investigations found a richer diversity of scaling behaviors of dynamic hysteresis [11,16,17].

For systems exhibiting FOPT but away from the thermodynamic limit and subject to finite fluctuations, fully characterizing the global landscape of hysteresis scaling across different timescales remains challenging. This difficulty arises from the complex interplay among multiple characteristic timescales and the inherent ambiguity in distinguishing between local and global equilibrium regimes [5,15,18,19]. Moreover, while dynamical phase transitions (DPT) inspired by fast-driving hysteresis phenomena have attracted significant attention [1,20–22], the transition regime from slow-driving FOPT to DPT has been largely overlooked, and whether a universal scaling exists in this intermediate regime remains an open question. We bridge these gaps in a companion paper [23] by proposing the concept of coercivity landscape. We have fully displayed the finite-time and finite-size scaling behaviors of hysteresis across multiple timescales and system sizes in the stochastic ϕ^4 model [8,17,24,25]. In this paper, we not only provide extensive analytical and numerical support for these results in a self-contained way, but also study the coercivity landscape of the Curie-Weiss model [15,26–28], clarifying universal scalings and model-specific scalings in the coercivity landscape.

*Contact author: xhzhao@mail.bnu.edu.cn

†Contact author: yhma@bnu.edu.cn

The rest of this paper is organized as follows. In Sec. II, we introduce the stochastic ϕ^4 model, specify its Langevin dynamics under a periodic driving field, and extract its hysteresis loops and coercivity from numerical simulations. In Sec. III, we derive the exact evolution equation for the ensemble-averaged order parameter, from which the finite-time scaling relations of coercivity in different timescales are obtained. Moreover, we investigate the finite-size scaling of the coercivity plateau by utilizing renormalization-group theory. To demonstrate the advantages of the coercivity landscape in studying hysteresis phenomena in concrete microscopic models, in Sec. IV, we investigate the magnetic hysteresis of the Curie-Weiss model. A universal collapse of coercivity across system sizes and model-specific finite-time scalings in the fast-driving regime are demonstrated. Conclusions are drawn in Sec. V.

II. STOCHASTIC ϕ^4 MODEL

The stochastic ϕ^4 model describes the dynamics of a system with a bistable potential subjected to external driving and thermal fluctuations. In the thermodynamic limit, the deterministic ϕ^4 model exhibits a sharp FOPT where the system's order parameter changes discontinuously as the external field crosses a critical value. To capture the finite-size effects, a noise term is introduced into the system's dynamics to represent finite fluctuations and the resulting stochastic behavior away from the thermodynamic limit [8,17,24,29–32]. By incorporating stochastic dynamics, the model allows for the exploration of dynamic hysteresis in finite systems while preserving the key features of first-order transitions, making it an ideal platform for understanding the interplay between finite-time driving and finite-size effects.

In this section, we investigate the dynamics of the stochastic ϕ^4 model under the driving of an external field and further demonstrate its hysteresis loops under periodic driving.

A. Dynamics

For a system of interest described by the mean-field Landau model interacting with an external field H , its free energy density, as a function of the dimensionless order parameter ϕ , follows [33]:

$$f_4(\phi, H) = \frac{1}{2}a_2\phi^2 + \frac{1}{4}a_4\phi^4 - H\phi. \quad (1)$$

Here the constants a_2 and a_4 correspond to the reduced temperature and intrinsic interaction strength, respectively. We focus on the case $a_2 < 0$, for which the system exhibits spontaneous symmetry breaking in the absence of both external field and fluctuations. The free energy landscape exhibits $\mathbb{Z}_2$ -symmetric bistability at zero field ($H = 0$). Under weak applied fields satisfying $|H| < |H_{\text{sp}}^{\pm}|$, this double-well potential becomes tilted while maintaining both local minima. By solving from $\partial_{\phi}f_4(\phi, H) = 0$ and $\partial_{\phi}^2f_4(\phi, H) = 0$, the spinodal field $H_{\text{sp}}^{\pm}$ is obtained as

$$H_{\text{sp}}^{\pm} = \pm H^* \equiv \pm \sqrt{(4/27)(-a_2^3/a_4)}, \quad (2)$$

at which the metastable state ceases to exist. This corresponds to a saddle-node bifurcation, where the metastable minimum

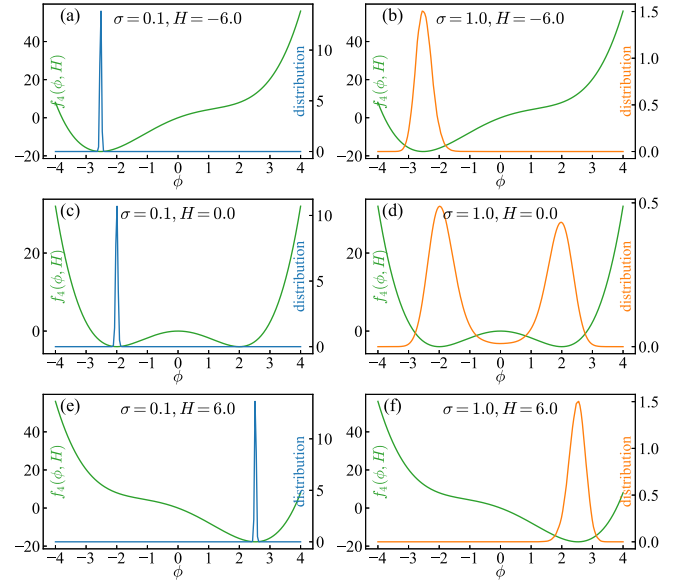


FIG. 1. Free-energy-density curves (with upturned ends) and probability-distribution curves (with flat ends) of the stochastic ϕ^4 model at different external field values H during a driven process. Panels [(a), (c), (e)] and [(b), (d), (f)] correspond to noise strengths $\sigma = 0.1$ and $\sigma = 1$, respectively. From top to bottom, $H = -6, 0, 6$. In this plot, the reduced temperature is $a_2 = -4$, the driving velocity is $v_H = 0.001$, and all quantities are nondimensionalized by the intrinsic parameters a_4 and λ .

merges with the unstable maximum at

$$\phi_{\text{sp}}^{\pm} = \pm \phi^* \equiv \mp \sqrt{-a_2/(3a_4)}, \quad (3)$$

where the superscripts $\pm$ denote transitions toward positive and negative ϕ , respectively. The dynamics of $\phi = \phi(t)$ under a time-dependent field $H = H(t)$ can be described by the Langevin equation [8,17,24]

$$\frac{\partial \phi}{\partial t} = -\lambda \frac{\partial f_4(\phi, H)}{\partial \phi} + \zeta(t). \quad (4)$$

Here λ sets a reference timescale and $\zeta(t)$ is a Gaussian white noise with amplitude $\sqrt{2\lambda\sigma^2}$ satisfying $\langle \zeta(t) \rangle = 0$ and $\langle \zeta(t)\zeta(t') \rangle = 2\lambda\sigma^2\delta(t-t')$, and $\sigma \geq 0$ denoting the noise strength.

Substituting the free energy expression Eq. (1) into the dynamic equation Eq. (4), the specific form of the dynamic equation reads

$$\frac{\partial \phi}{\partial t} = -\lambda a_2 \phi - \lambda a_4 \phi^3 + \lambda H + \zeta(t). \quad (5)$$

The change of the order parameter over a small time interval δt is given by the discretized form of the above equation [34],

$$\delta \phi = -\lambda a_2 \phi \delta t - \lambda a_4 \phi^3 \delta t + \lambda H \delta t + \sqrt{2\lambda\sigma^2 \delta t} W, \quad (6)$$

where W is a random variable drawn from a standard normal distribution. By simulating Eq. (6), we can obtain the evolution of $\phi(t)$ under varying field $H(t)$. Figure 1 shows the potential curves and corresponding probability distributions in increasing-field processes for two different noise strengths. From top to bottom, the field is strongly negative, zero, and strongly positive, respectively. For the smaller noise

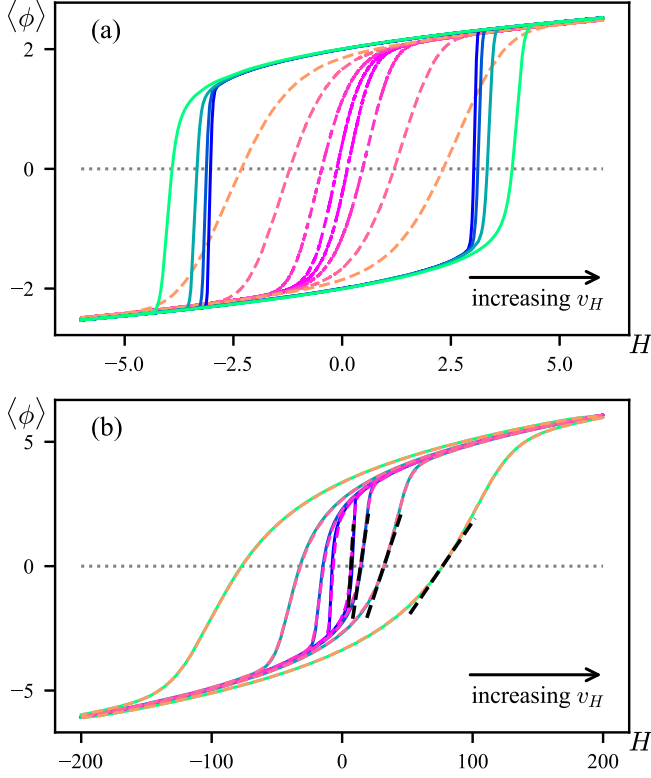


FIG. 2. Hysteresis loops for $\sigma = 0.1$ (solid curves) and $\sigma = 1.0$ (dashed curves). (a) Slow driving, $v_H = 0.004, 0.02, 0.1, 0.5$ from the inside out. (b) Fast driving, $v_H = 8, 40, 200, 1000$ from the inside out. $a_2 = -4$ and all quantities nondimensionalized by a_4 and λ .

strength, the probability distribution remains highly concentrated around the initial minimum. The system retains its initial state well past the field where the two local minima become energetically symmetric. In contrast, for the larger noise strength, the wider probability distribution facilitates an earlier transition from the initial state to the final equilibrium state due to enhanced fluctuations.

B. Hysteresis loops

We control the external field varying linearly between $\mp H_m$ at a rate $|dH/dt| = \lambda v_H$, where $H_m > 0$ is the amplitude. With the system initially prepared in the equilibrium state corresponding to the strongly negative field $-H_m$, the averaged order parameter $\langle \phi \rangle$ evolves from a negative to a positive value and then returns to the negative one. As the external field changes, the system responds by adjusting $\langle \phi \rangle$, which reflects the competition between the driving field and the system's internal dynamics. The point at which $\langle \phi \rangle$ crosses zero defines the coercivity, marking the field strength at which the system's response changes direction.

Figures 2(a) and 2(b) show the saturated hysteresis loops in different driving regimes. Here, “Saturated” or “saturation” refers to hysteresis loops for which the irreversibility (loop area) no longer increases upon further increasing the field amplitude. Operationally, under periodic driving starting from an ordered state, a saturated hysteresis loop becomes stable after one driving period, whereas an unsaturated loop remains in a transient state during the early driving periods. The coercivity,

as reflected in the $\langle \phi \rangle - H$ curves, increases with the driving velocity. As illustrated in Fig. 2(a), the curves in the relatively slow driving regime exhibit distinct behaviors depending on the noise strength: for large noise strength associated with the dashed loops, the transitions are smoother and the coercivity is smaller; for small noise strength associated with the solid loops, the transitions are sharper and the coercivity is larger. This difference arises from the noise-induced reduction in relaxation time. As the external field increases and the left potential well becomes metastable, the escape rate to the stable well is enhanced by larger noise strength, which can be estimated by the Kramers' escape rate $\sim \exp(-\Delta E/\sigma^2)$, where ΔE is the height of the energy barrier [35]. As a result, the order parameter transitions earlier, at a smaller field value. In the relatively fast driving regime, the $\langle \phi \rangle - H$ curves plotted in Fig. 2(b) for different noise strengths nearly overlap. This is because the driving velocity exceeds any intrinsic relaxation timescale of the system, regardless of the noise strength, leading to similar forced relaxation processes.

III. COERCIVITY OF THE STOCHASTIC ϕ^4 MODEL

Coercivity is a fundamental property of the hysteresis loop, especially for magnetic and ferroelectric materials, defining the external magnetic or electric field strength required to reduce a material's magnetization or polarization to zero. It indicates how well a material resists demagnetization or depolarization: high coercivity materials are “hard” and maintain their state effectively, ideal for permanent magnets in devices like electric motors [36], while low coercivity materials are “soft” and easily switched, commonly used in transformers [37]. This property is vital for optimizing performance in technologies such as magnetic storage devices and sensors [38,39]. At a general level, in our current work, we define the coercivity $H_c \equiv H(\langle \phi \rangle = 0)$ as a mesoscopic or macroscopic observable of the hysteresis phenomenon caused by microscopic dynamics. Clarifying the change of coercivity with the external driving is of great significance for theoretically quantifying the hysteresis phenomenon and engineering the performance of devices constructed with hysteretic materials in practical applications. Below, we provide a systematic exploration of the coercivity landscape proposed in the companion paper [23].

A. Coercivity landscape

In [Fig. 1(e)] of the companion paper [23], we presented a coercivity landscape. Here, a more comprehensive coercivity landscape is provided in Fig. 3. In this plot, for simulation simplicity, we adopt monotonically increasing-field driving instead of periodic driving. The initial field magnitude is chosen slightly larger than the spinodal field for the given a_2 and a_4 , such that the free-energy landscape has a single stable minimum, and the system is prepared in the corresponding equilibrium state. In Fig. 3(a), the curves from top to bottom correspond to increasing noise strength σ . For the largest noise strength, we include two reference lines in the fast- and slow-driving regimes, indicating power-law behaviors with exponents $1/2$ and 1 , respectively. For the largest noise strength associated with the topmost curve in Fig. 3(b) the scaling exponent increases monotonically from $1/2$ to 1 as

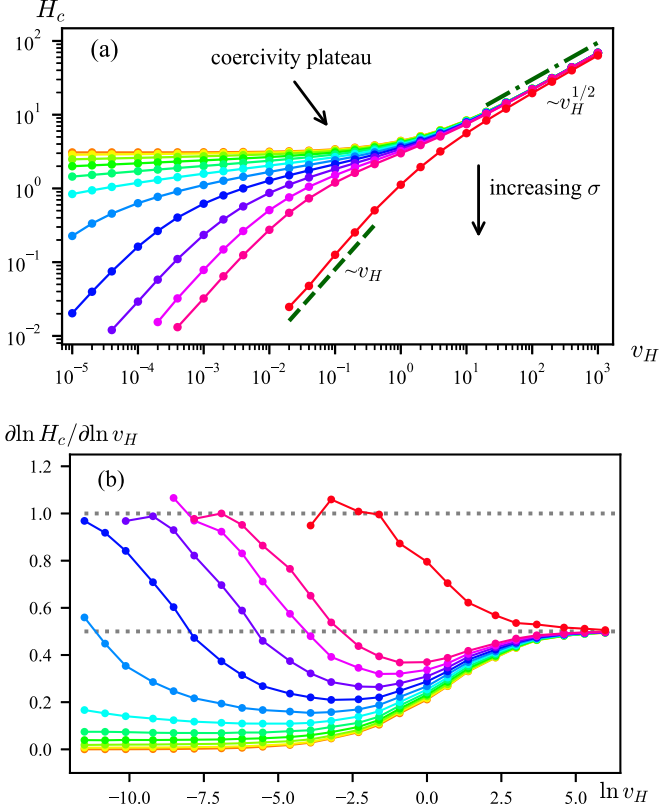


FIG. 3. (a) Coercivity values H_c and (b) their local scaling exponents $\partial \ln H_c / \partial \ln v_H$ with respect to the driving rate v_H under the saturation field condition for different noise strengths. $\sigma = 0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 2$ from top to bottom in panel (a) while from bottom to top in panel (b). In this plot, $a_2 = -4$, $a_4 = 1$, $\lambda = 1$.

the velocity decreases. However, for smaller noise strengths, the exponent first decreases before increasing toward 1 in the very slow-driving regime. When the noise strength is very small, the exponent continues to decrease throughout the simulated velocity range, although it is expected to eventually increase at even slower driving. The decreasing behavior in the scaling exponent reflects a plateau-like region in the coercivity-velocity curves, where the coercivity changes very slowly with velocity. The plateau-like behavior of the coercivity has not been reported in previous studies. We interpret this coercivity plateau as a characteristic feature of a finite-sized ergodicity-breaking effect with the system localized in the metastable state. It becomes increasingly broad and pronounced as the noise strength decreases and, in the thermodynamic limit, it is expected to evolve into an asymptotic plateau, indicating an exact FOPT that is triggered at a finite coercivity value in the quasistatic limit. The specific coercivity plateau point is defined as the local minimum point shown in Fig. 3(b), while the plateau point for very small noise strength has not been achieved in the presented simulation timescale.

To explicitly understand the response of coercivity with respect to the driving rate, Eq. (4) is equivalently expressed by the following Fokker-Planck equation,

$$\frac{\partial P(\phi, t)}{\partial t} = \lambda \frac{\partial}{\partial \phi} \left[\frac{\partial f_4(\phi, H)}{\partial \phi} + \sigma^2 \frac{\partial}{\partial \phi} \right] P(\phi, t), \quad (7)$$

where $P(\phi, t)$ is the probability density function. From the above equation, we obtain the evolution of $\langle \phi \rangle$ in time,

$$v_H \frac{d\langle \phi \rangle}{dH} = -a_2 \langle \phi \rangle - a_4 \langle \phi^3 \rangle + H, \quad (8)$$

where we have used the linear variation of $H(t)$, that is $H(t) = H_0 + \lambda v_H t$, and hence $dH = \lambda v_H dt$. Hereafter, the increasing branch of the field function is considered in analytical calculations, unless specifically stated otherwise. The second term on the right-hand side of the above equation may be expressed as

$$\langle \phi^3 \rangle = \langle \phi \rangle^3 + 3\mu_2 \langle \phi \rangle + \mu_3, \quad (9)$$

where $\mu_2 \equiv \langle (\phi - \langle \phi \rangle)^2 \rangle$, $\mu_3 \equiv \langle (\phi - \langle \phi \rangle)^3 \rangle$ are the second and third central moments of the random variable ϕ , respectively. In particular, μ_3 is associated with the skewness of the distribution of ϕ . The coercivity is expressed as

$$H_c = v_H \left. \frac{d\langle \phi \rangle}{dH} \right|_{\langle \phi \rangle=0} + a_4 \mu_3 \Big|_{\langle \phi \rangle=0}. \quad (10)$$

B. $H_c \sim v_H$ scaling in the near equilibrium regime

Linear scaling is observed in the near-equilibrium regime. In the equilibrium limit, the order parameter is distributed symmetrically in the double-well potential landscape, and therefore $\langle \phi \rangle$ becomes zero at $H = 0$, indicating that the coercivity vanishes in this limit. Due to this symmetric distribution, the skewness of the equilibrium distribution, represented by the third central moment μ_3 , also vanishes. In one word, both the equilibrium-limit values of H_c and μ_3 at $\langle \phi \rangle = 0$ are zero.

To quantify deviations from equilibrium in the near-equilibrium regime, the key quantities, H_c , $\mu_3|_{\langle \phi \rangle=0}$, and $(d\langle \phi \rangle/dH)|_{\langle \phi \rangle=0}$, in Eq. (10) can be respectively expanded in power series around $v_H = 0$,

$$H_c = h_1 v_H^\alpha + h_2 v_H^{2\alpha} + \dots, \quad (11a)$$

$$\mu_3 = u_1 v_H^\beta + u_2 v_H^{2\beta} + \dots, \quad (11b)$$

$$\left. \frac{d\langle \phi \rangle}{dH} \right|_{\langle \phi \rangle=0} = \left. \frac{d\langle \phi \rangle_{\text{eq}}}{dH} \right|_{\langle \phi \rangle_{\text{eq}}=0} + g_1 v_H^\gamma + g_2 v_H^{2\gamma} + \dots, \quad (11c)$$

where the subscript eq denotes equilibrium limit. Substituting the above series into Eq. (10) and retaining of the lowest-order corrections in each term, we find that $\alpha = \beta = 1$ to balance the powers of v_H . Therefore, the near-equilibrium correction of the coercivity follows as

$$H_c \sim v_H. \quad (12)$$

We emphasize that the above analysis applies to finite noise strength, as $(d\langle \phi \rangle/dH)|_{\langle \phi \rangle=0}$ diverges in the equilibrium limit when the noise vanishes, and thus cannot be expanded in a power series like Eq. (11c) near equilibrium.

C. $H_c \sim v_H^{1/2}$ scaling in fast-driving regime

In the fast-driving regime, coercivity shows no significant dependence on noise strengths. This is because the external field changes so rapidly that the system remains effectively frozen before the potential landscape becomes monostable.

In this regime, the dynamics are dominated by deterministic forces, while fluctuations, which are crucial for transitions between metastable states, play a negligible role. Consequently, the coercivity behavior can be understood by analyzing the relaxation dynamics toward the monostable state in systems with vanishing fluctuations under a time-dependent strong field.

For a varying strong field, the position of the monostable state, that is the equilibrium state, in the ϕ^4 model also changes accordingly. When the magnitude of the field is sufficiently large, specifically for $|H| \gg \sqrt{|a_2^3/a_4|}$, the equilibrium state can be approximated as follows:

$$\phi_{\text{eq}} = \left(\frac{H}{a_4}\right)^{1/3}. \quad (13)$$

In deriving the above expression, the $a_2\phi$ term in $\partial f_4(\phi, H)/\partial\phi$ has been neglected, as it becomes negligible compared to the $a_4\phi^3$ term when the external field H is sufficiently large.

The relaxation dynamics near the equilibrium state can be approximated by the following linearized equation,

$$\frac{\partial \langle\phi\rangle}{\partial t} \approx -\lambda \frac{\partial^2 f_4(\phi, H)}{\partial \phi^2} \Big|_{\phi_{\text{eq}}} (\langle\phi\rangle - \phi_{\text{eq}}), \quad (14)$$

from which we obtain that the characteristic relaxation time to the equilibrium state is

$$\begin{aligned} \tau_{\text{relax}} &= \frac{1}{\lambda(\partial^2 f_4/\partial \phi^2)|_{\phi_{\text{eq}}}} = \frac{1}{\lambda(a_2 + 3a_4\phi_{\text{eq}}^2)} \\ &\approx \frac{1}{3\lambda a_4\phi_{\text{eq}}^2} \\ &\approx \frac{1}{3\lambda a_4^{1/3} H^{2/3}}, \end{aligned} \quad (15)$$

where we have used the expression of ϕ_{eq} .

With an increasing external field, the equilibrium state moves with the rate

$$r_{\phi_{\text{eq}}} \equiv \frac{d\phi_{\text{eq}}}{dH} \lambda v_H \approx \frac{1}{3} \frac{\lambda v_H}{a_4^{1/3} H^{2/3}}. \quad (16)$$

The characteristic rate of relaxation is

$$r_{\text{relax}} \equiv \frac{1}{\tau_{\text{relax}}} \approx 3\lambda a_4^{1/3} H^{2/3}. \quad (17)$$

Since the characteristic rate of relaxation increases and the moving rate of the equilibrium state decreases with the external field for fixed v_H , we identify that the equation

$$r_{\text{relax}} \simeq r_{\phi_{\text{eq}}} \quad (18)$$

represents a turning point of the field that the relaxation of the system becomes able to catch up the variation of the equilibrium state and we define this turning point as H_{sat} . By substituting the expressions of r_{relax} and $r_{\phi_{\text{eq}}}$ into the above equation, we obtain that H_{sat} scales with v_H as follows:

$$H_{\text{sat}} \sim v_H^{3/4}. \quad (19)$$

To associate H_{sat} with the interested coercivity, we introduce a reference right triangle on the hysteresis curve as

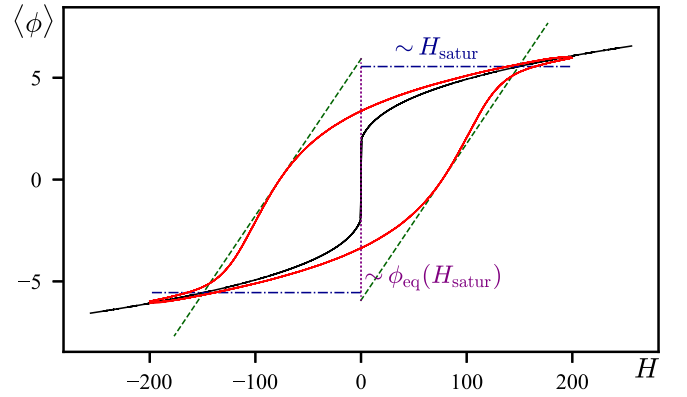


FIG. 4. Geometric illustration of the hysteresis loop catching up to the equilibrium in the fast-driving regime. $\sigma = 0.7$, $v_H = 1000$ are used to generate the hysteresis loop while the central transition curve represents the equilibrium limit. In this plot, $a_2 = -4$, $a_4 = 1$, $\lambda = 1$.

illustrated in Fig. 4. The hypotenuse of the triangle is defined by the line passing through the coercivity point with the same slope as the $\langle\phi\rangle$ - H curve at this point, and it intersects the equilibrium curve as one vertex of the triangle. The vertical leg of the triangle is defined to pass through the origin, and the horizontal leg is determined accordingly.

Since the averaged order parameter value approaches the equilibrium value near the right terminal point of the horizontal leg, the length of the horizontal leg is approximated by the magnitude of H_{sat} . Accordingly, the length of the vertical leg is approximated by $\phi_{\text{eq}}(H_{\text{sat}})$. Then the slope of the hypotenuse is approximated by $\phi_{\text{eq}}(H_{\text{sat}})/H_{\text{sat}}$, which gives the following scaling relation with respect to v_H ,

$$\frac{d\langle\phi\rangle}{dH} \Big|_{\langle\phi\rangle=0} \sim H_{\text{sat}}^{-2/3} \sim v_H^{-1/2}. \quad (20)$$

Furthermore, in the fast driving regime, the skewness of ϕ is negligible in Eq. (10), and hence the coercivity is approximated by

$$H_c = v_H \frac{d\langle\phi\rangle}{dH} \Big|_{\langle\phi\rangle=0}, \quad (21)$$

which is verified in Fig. 2(b), where the tilted dashed line is plotted with the slope H_c/v_H . Hence, the scaling behavior of the coercivity with respect to driving velocity in the fast driving regime is

$$H_c \sim v_H^{1/2}, \quad (22)$$

which is universal for the stochastic ϕ^4 model. This universality holds not only for different σ [see Fig. 3(a)] but also for different a_2 and a_4 ; see Appendix A for details.

We emphasize here that the $v_H^{2/3}$ scaling proposed as a complete-scaling behavior in Ref. [40] does not conflict with our findings. Specifically, our derived scalings, $v_H^{1/2}$ in the fast driving regime and v_H in the near equilibrium regime (Sec. III B), are results associated with fixed system parameters. These two regimes can be covered by the complete scaling with covarying parameters. A detailed analysis of this consistency is provided in Appendix B.

D. Finite-size scaling of the coercivity plateau

In this subsection, we examine how finite-size effects influence the coercivity plateau, which is a characteristic signature of effective dynamical ergodicity breaking in finite-time FOPT. In particular, we focus on how the plateau height and its location scale with the noise strength. Although noise is often associated with finite temperature, within our framework we focus on interpreting the noise parameter σ as finite-size effects, reflecting the fact that environment-induced relative fluctuations become increasingly significant as the system size decreases [29–32]. In our mean-field treatment of interest, finite-size and finite-noise effects are further intertwined through their analogous roles in the Kramers escape rate, specifically via the effective potential barrier [5].

Since the plateau is associated with dynamical ergodicity breaking in systems with small noise, we will expand the quartic potential near the field-driven FOPT point in the thermodynamic limit, retaining terms up to the cubic order of the order parameter [3,40]. This is justified by the fact that the order parameter undergoes an asymmetric discontinuous flip during FOPT. The transition point of the system in the thermodynamic limit is exactly the spinodal point (H^*, ϕ^*) given by Eqs. (2) and (3). Defining $h \equiv H - H^*$ and $\varphi \equiv \phi - \phi^*$, the original quartic potential (1) is expanded as

$$f_3(\varphi, h) = \frac{1}{3}a_3\varphi^3 - h\varphi, \quad (23)$$

where $a_3 = -\sqrt{-3a_2a_4}$. Substitute the above cubic potential into the governing Langevin equation (4) and then one obtains the corresponding discretized equation as

$$\delta\varphi = -a_3\varphi^2v_H^{-1}\delta h + hv_H^{-1}\delta h + \sqrt{2\sigma^2v_H^{-1}}\delta hW, \quad (24)$$

where we have used the relation $\delta h = \lambda v_H \delta t$ for linearly increasing field. In Ref. [40], Zhong applies scaling transformations to the dynamic equation governed by the cubic potential, based on renormalization-group theory, and derives complete scaling relations for all system parameters and variables with respect to the driving velocity. In the present work, to investigate the finite-size effect on the coercivity plateau, we perform analogous scaling transformations with respect to the noise strength. Using Σ as the scaling factor to scale the noise strength from σ to $\sigma' = \sigma \Sigma^{-1}$, the quantity $\mathcal{O}$ becomes $\mathcal{O}' = \mathcal{O}\Sigma^{-[\mathcal{O}]}$, where $[\mathcal{O}]$ is the scaling dimension of $\mathcal{O}$. Performing scaling transformations on Eq. (24), one has

$$\begin{aligned} \delta\varphi' \Sigma^{[\varphi]} &= -a'_3\varphi'^2v_H'^{-1}\delta h' \Sigma^{[a_3]+2[\varphi]-[v_H]+[h]} \\ &\quad + h'v_H'^{-1}\delta h' \Sigma^{2[h]-[v_H]} \\ &\quad + \sqrt{2\sigma'^2v_H'^{-1}}\delta h'W \Sigma^{1-[v_H]/2+[h]/2}. \end{aligned} \quad (25)$$

To keep the dynamic equation invariant after scaling transformation, the scaling dimensions of each term in the equation should balance, that is

$$\begin{aligned} [\varphi] &= [a_3] + 2[\varphi] - [v_H] + [h] = 2[h] - [v_H] \\ &= 1 - [v_H]/2 + [h]/2. \end{aligned} \quad (26)$$

According to renormalization-group theory, the scaling dimension of the coefficient in the highest-order term is set to be zero [40], that is $[a_3] = 0$. Substituting $[a_3] = 0$ into Eq. (26),

we obtain that

$$[h] = \frac{4}{3}, \quad [\varphi] = \frac{2}{3}, \quad [v_H] = 2. \quad (27)$$

The above scaling dimensions reveal that the scaling exponents of h , φ and v_H with respect to σ are respectively $4/3$, $2/3$ and 2 . These scaling exponents apply to cases where the hysteresis curves are adjacent to the quasistatic phase-transition curve in the deterministic limit. For finite σ , this corresponds to the coercivity plateau regime. Although these scaling relations are directly applicable to switching points analogous to the spinodal point within the hysteresis curves, it is verified that coercivity points also exhibit these scaling relations due to the small difference between switching points and coercivity points in abrupt transition processes. Therefore, we conclude the scaling exponents as

$$|H_P - H^*| \sim \sigma^{4/3} \quad \text{at} \quad v_P \sim \sigma^2, \quad (28)$$

where H_P is the height of the coercivity plateau, $|H_P - H^*|$ is the difference in the height between systems with finite and vanishing noise strengths and v_P is the reference driving rate at which the coercivity plateau is observed. Considering that the plateau height decreases with increasing σ , as shown in Fig. 3(a), we rewrite the above relations as

$$H^* - H_P \sim \sigma^{4/3} \quad \text{at} \quad v_P \sim \sigma^2. \quad (29)$$

This finite-size scaling is numerically validated in [Fig. 2(a)] of the companion paper [23].

In addition to the scaling dimensions for h , φ and v_H shown in Eq. (27), we further derive those for a_2 and a_4 , which together form a complete scaling set with respect to σ . From the relation $a_3 = -\sqrt{-3a_2a_4}$, it follows that $[a_2] + [a_4] = 2[a_3] = 0$. Assuming scaling for either the reduced field $[H^*] = [h] = [H]$ (which yields $3[a_2]/2 - [a_4]/2 = [h] = 4/3$), or for the reduced order parameter, $[\phi^*] = [\varphi] = [\phi]$ (which yields $[a_2]/2 - [a_4]/2 = [\varphi] = 2/3$), we finally obtain $[a_2] = 2/3$ and $[a_4] = -2/3$. When a_2 , a_4 , and v_H covary according to this complete scaling under different σ , the resulting hysteresis curves collapse to a unique one under rescaled H and ϕ as demonstrated in Figs. 5(a) and 5(b). Moreover, the coercivity curves under covarying parameters also collapse, as shown in Figs. 5(c) and 5(d), which further demonstrates the finite-size scaling relations obtained near the coercivity plateau.

E. $H_c - H_P \sim (v_H - v_P)^{2/3}$ scaling in the postplateau regime

Building on the characterization of the coercivity plateau, we now analyze the finite-time scaling in the postplateau. According to Eq. (8), for a given initial condition, the hysteresis curve under linearly varying field is uniquely determined by the driving rate v_H . Considering the specific hysteresis curve corresponding to $v_P(\sigma)$ for some σ , denote the switching point that triggers a rapid transition within this curve as (H_P^*, ϕ_P^*) . Expanding the free energy around this switching point and retaining to the third-order term, we have

$$f_3(\varphi_P, h_P) = b_0 + b_1\varphi_P + \frac{1}{2}b_2\varphi_P^2 + \frac{1}{3}b_3\varphi_P^3 - \phi_P^*h_P - h_P\varphi_P, \quad (30)$$

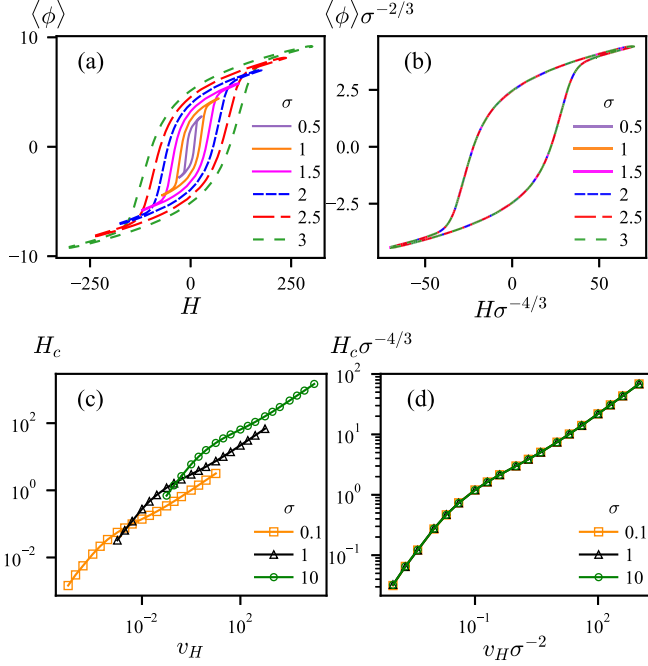


FIG. 5. Collapse of (a), (b) hysteresis loops and (c), (d) coercivity curves under complete scaling with respect to the noise strength. Different curves are generated according to $a_2 = a_2^{\text{ref}}(\sigma/\sigma^{\text{ref}})^{2/3}$, $a_4 = a_4^{\text{ref}}(\sigma/\sigma^{\text{ref}})^{-2/3}$ and $v_H = v_H^{\text{ref}}(\sigma/\sigma^{\text{ref}})^2$. $\sigma^{\text{ref}} = 1.0$, $a_2^{\text{ref}} = -4.0$ and $a_4^{\text{ref}} = 1.0$. In panels (a), (b), $v_H^{\text{ref}} = 100$. In panels (c), (d), $v_H^{\text{ref}} \in [10^{-3}, 10^3]$. In this plot, $\lambda = 1$.

where $\varphi_P \equiv \phi - \phi_P^*$, $h_P \equiv H - H_P^*$, representing the delay of the switching point when $v_H > v_P$, and

$$\begin{aligned} b_0 &= \frac{1}{4}a_4\phi_P^{*4} + \frac{1}{2}a_2\phi_P^{*2} - H_P^*\phi_P^*, \\ b_1 &= a_4\phi_P^{*3} + a_2\phi_P^* - H_P^*, \\ b_2 &= 3a_4\phi_P^{*2} + a_2, \\ b_3 &= 3a_4\phi_P^*. \end{aligned}$$

Substituting Eq. (30) into the governing Fokker-Planck equation (7),

$$\begin{aligned} \frac{\partial P(\varphi_P, t)}{\partial t} &= \lambda \frac{\partial}{\partial \varphi_P} [(b_1 + b_2\varphi_P + b_3\varphi_P^2 - h_P)P(\varphi_P, t)] \\ &\quad + \lambda \sigma^2 \frac{\partial^2 P(\varphi_P, t)}{\partial \varphi_P^2}. \end{aligned} \quad (31)$$

Multiplying both sides of the above equation by φ_P and integrating over φ_P , one obtains the averaged equation

$$v_H \frac{d\langle \varphi_P \rangle}{dh_P} = -b_1 - b_2\langle \varphi_P \rangle - b_3\langle \varphi_P^2 \rangle + h_P, \quad (32)$$

where the relation $dh_P = \lambda v_H dt$ has been used. Evaluating the above equation at $\langle \varphi_P \rangle = 0$, the result reads

$$v_H \left. \frac{d\langle \varphi_P \rangle}{dh_P} \right|_{\langle \varphi_P \rangle=0} = -b_1 - b_3\langle \varphi_P^2 \rangle \Big|_{\langle \varphi_P \rangle=0} + h_P \Big|_{\langle \varphi_P \rangle=0}. \quad (33)$$

For $v_H = v_P$, the above relation becomes

$$v_P \left. \frac{d\langle \varphi_P \rangle}{dh_P} \right|_{\langle \varphi_P \rangle=0, v_P} = -b_1 - b_3\langle \varphi_P^2 \rangle \Big|_{\langle \varphi_P \rangle=0, v_P}. \quad (34)$$

Defining $v_h \equiv v_H - v_P$ and expanding Eq. (33) around $v_h = 0$, we obtain

$$v_h s = -b_3\mu + h_P \Big|_{\langle \varphi_P \rangle=0}, \quad (35)$$

where

$$\begin{aligned} s &\equiv \left. \frac{d\langle \varphi_P \rangle}{dh_P} \right|_{\langle \varphi_P \rangle=0} - \left. \frac{d\langle \varphi_P \rangle}{dh_P} \right|_{\langle \varphi_P \rangle=0, v_P}, \\ \mu &\equiv \langle \varphi_P^2 \rangle \Big|_{\langle \varphi_P \rangle=0} - \langle \varphi_P^2 \rangle \Big|_{\langle \varphi_P \rangle=0, v_P}. \end{aligned} \quad (36)$$

Performing scaling transformations on Eq. (35) with respect to v_h , which gives $[v_h] = 1$, $[s] \sim [\varphi_P] - [h_P]$, $[\mu] \sim 2[\varphi_P]$, and setting $[b_3] = 0$, the resulting balanced scaling relation is $h_P|_{\langle \varphi_P \rangle=0} \sim v_h^{2/3}$. Due to the closeness between the switching point and the coercivity point for hysteresis loops with a feature of first-order phase transition, the coercivity value also adheres to this scaling, that is

$$H_c - H_P \sim (v_H - v_P)^{2/3}. \quad (37)$$

IV. COERCIVITY LANDSCAPE OF THE CURIE-WEISS MODEL

In this section, we apply the coercivity landscape approach to study a specific microscopic model, the Curie-Weiss model [15,26–28]. We will extend the discussion on the stochastic ϕ^4 model to systematically study the finite-time and finite-size scaling of the coercivity in the magnetic hysteresis of the Curie-Weiss model. Furthermore, the universal scalings and model-dependent scalings in the coercivity landscape will be clarified.

A. Dynamics and hysteresis

The Curie-Weiss model is a paradigmatic mean-field model for ferromagnetic systems, capturing phase transitions through infinite-range interactions. Consider a system of N Ising spins $\sigma_i = \pm 1$, labeled $i = 1, \dots, N$, where each spin interacts with all others via a ferromagnetic coupling $J/2N$, and define the total magnetization $M = \sum_{i=1}^N \sigma_i$. The system is immersed in a thermal bath at inverse temperature $\beta = 1/(k_B T)$ and driven by an external time-dependent magnetic field $H(t)$. The Hamiltonian is given by [15,26]

$$\mathcal{H} = -\frac{J}{2N}M^2 - H(t)M. \quad (38)$$

When an arbitrary spin flips from ∓ 1 to ± 1 , the total magnetization will change by two units, i.e., $M \rightarrow M_{\pm} \equiv M \pm 2$. The probability $P(M, t)$ of observing total magnetization M at time t is governed by the master equation

$$\begin{aligned} \frac{dP(M, t)}{dt} &= \sum_{\eta=\pm} [W_{\eta}(M_{-\eta}, H)P(M_{-\eta}, t) \\ &\quad - W_{\eta}(M, H)P(M, t)], \end{aligned} \quad (39)$$

with the transition rates for $M \rightarrow M_{\pm} \equiv M \pm 2$,

$$W_{\pm}(M, H) = \frac{N \mp M}{2\tau_0} \exp \left[\pm \beta \left(J \frac{M \pm 1}{N} + H \right) \right], \quad (40)$$

and the microscopic relaxation time τ_0 . These transition rates satisfy the detailed balance condition

$$W_{\pm}(M, H)P_{\text{eq}}(M, H) = W_{\mp}(M_{\pm}, H)P_{\text{eq}}(M_{\pm}, H), \quad (41)$$

where $P_{\text{eq}}(M, H) \propto \exp[-\beta \mathcal{F}_H(M)]$ is the equilibrium distribution with the free energy

$$\mathcal{F}_H(M) = -\frac{J}{2N}M^2 - HM - \frac{1}{\beta} \ln \Omega(M), \quad (42)$$

and $\Omega(M) = N!/[(N+M)/2]![(N-M)/2]!$ is the microscopic degeneracy of magnetization M .

By multiplying both sides of Eq. (39) by M and sum over all possible states, the time evolution of the average magnetization $\langle M \rangle$ is given by

$$\begin{aligned} \frac{d\langle M \rangle}{dt} = & \sum_M \sum_{\eta=\pm} [MW_{\eta}(M_{-\eta}, H)P(M_{-\eta}, t) \\ & - MW_{\eta}(M, H)P(M, t)], \end{aligned} \quad (43)$$

which, in the thermodynamic limit of $N \rightarrow \infty$, becomes the deterministic evolution equation of $m \equiv M/N$ [15,27]

$$\begin{aligned} \frac{dm}{dt} = & \frac{2}{\tau_0} [\sinh(\beta Jm + \beta H) \\ & - m \cosh(\beta Jm + \beta H)] \equiv g(m, H). \end{aligned} \quad (44)$$

By solving this equation with a linearly driving protocol with an amplitude H_m and a rate of $|dH/dt| = v_H$, we obtain the hysteresis loop for the infinite system ($N \rightarrow \infty$) as the dotted curve in Fig. 6(a) with $v_H = 0.001$. In this example, we use $\tau_0 = 2$, $\beta = 2$, and $J = 1$. Unless otherwise stated, the parameters used in the other figures of this section are the same as those here.

To illustrate the finite-size effect on hysteresis loop, we numerically solve the master equation Eq. (39) to obtain the probability distribution $P(M, t)$, and compute the per-site average magnetization $\langle m \rangle = [\sum_M MP(M, t)]/N$, for $N = 30, 90, 150$. As shown in Fig. 6(a), increasing the system size N sharpens the loops, since thermal fluctuations are progressively suppressed, and the curves approach the deterministic, almost rectangular shape expected in the thermodynamic limit. Additionally, in Fig. 6(b), with fixed $N = 30$ and varied field amplitude H_m , larger H_m results in wider loops and higher coercivity, while small H_m leads to unsaturated, narrow and flat hysteresis loops. This trend arises because only strong fields enable spins to overcome energy barriers for complete state transitions.

Next, we will mainly focus on how the driving rate sculpts both loop morphology and coercivity at a fixed amplitude. In the slow-driving regime Fig. 7(a), the curves remain nearly centrally symmetric with respect to the origin, opening gradually as v_H increases. As the system enters the fast-driving regime Fig. 7(b), hysteresis loops become increasingly tilted and asymmetric, marking the onset of DPT. Figure 7(c) exhibits the coercivity landscape as a function of v_H , linking loop morphology across all rates. This landscape strikingly

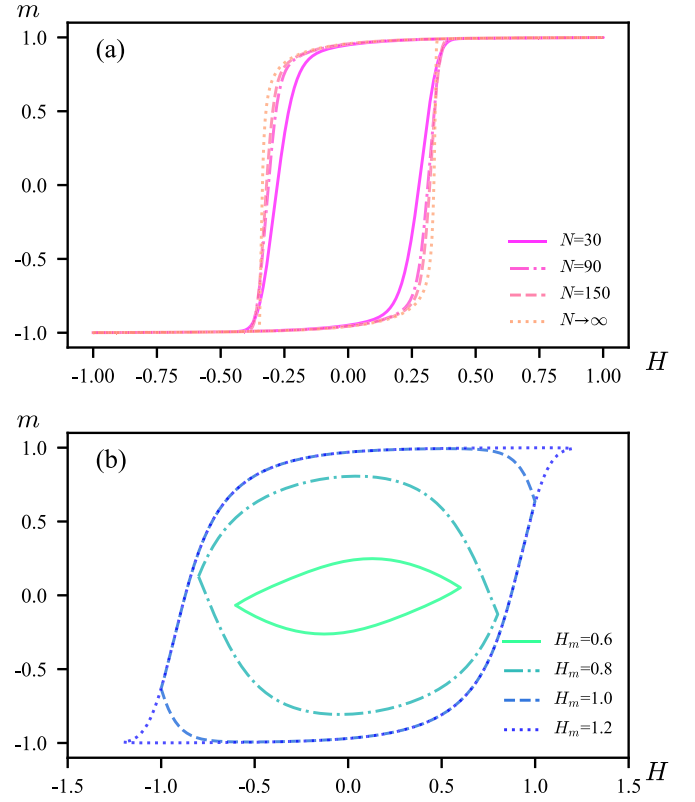


FIG. 6. Hysteresis loops of the Curie-Weiss model under periodic driving for (a) different spin number N at fixed amplitude $H_m = 1$; (b) different driving amplitude H_m with $N = 30$. In this plot, the parameters $\tau_0 = 2$, $\beta = 2$, and $J = 1$ are used. Unless otherwise stated, the parameters used in the other figures of this section are the same as those here.

reveals four distinct dynamic regimes, analogous to those of the stochastic ϕ^4 model in [23]: (i) a near-equilibrium regime where coercivity scales linearly with v_H ; (ii) a coercivity-plateau regime with effective ergodicity breaking; (iii) a postplateau power-law scaling regime; and (iv) a DPT regime with coercivity rapidly diminishing at high driving rates. Regimes (i)–(iii) all correspond to phase-to-phase switching. The v_H scaling in the near-equilibrium regime and the emergence of both the coercivity plateau and the diminishing DPT regime are all universal across both the stochastic ϕ^4 and Curie-Weiss models. It should be noted, however, that the power-law scaling after the plateau exhibits an exponent of $1/3$ that differs from the $1/2$ obtained in the stochastic ϕ^4 model, which will be calculated in detail in the next subsection. Building on the above single-size landscape, Fig. 8 extends this investigation by incorporating multiple system sizes N under the same parameter settings, thereby illustrating how system size controls the coercivity landscape. At the small rate range of the figure, curves of small systems from 5 to 30, collapse onto the near-equilibrium linear regime ($H_c \sim v_H$). For larger systems, smaller driving speeds (less than 10^{-6}) are required to observe this scaling. As v_H increases, a plateau appears where H_c remains relatively constant; the width of this plateau grows with N , and its onset shifts to lower v_H as finite-size effect diminishes.

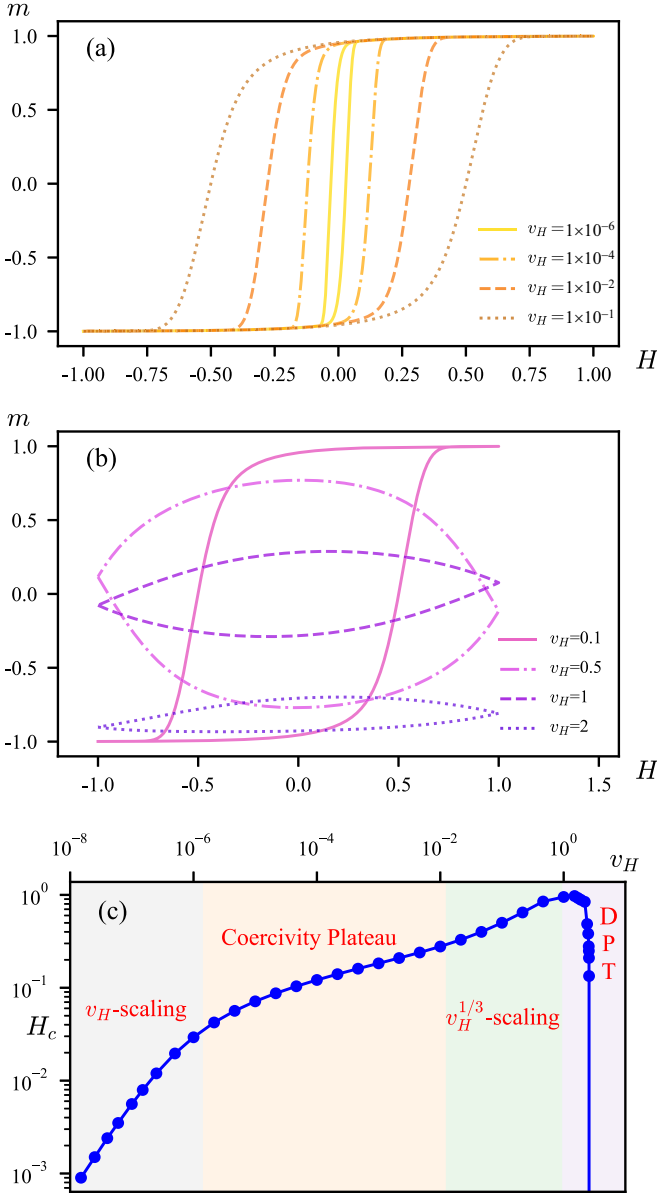


FIG. 7. Hysteresis loops and coercivity landscape of the Curie-Weiss model with $N = 30$ and $H_m = 1$. (a) Slow-driving loops go through the near-equilibrium regime, the coercivity plateau and the postplateau. (b) Fast-driving loops transition from the postplateau to the DPT (dynamical phase transition) regime. (c) Coercivity H_c versus driving rate v_H on a log-log scale.

Immediately following, at further higher rates, all curves converge onto the fast-driving asymptote ($\sim v_H^{1/3}$). Ultimately, all system sizes transition into the DPT regime. The larger the system size, the faster the coercivity decreases to zero. The behaviors described above are obtained under the condition of controlled field amplitude H_m , where the hysteresis loops do not necessarily reach full saturation. In contrast, if the external field amplitude H_m is not constrained and the hysteresis loops are driven into full saturation, additional scaling behaviors emerge at high driving rate, which will be analyzed later.

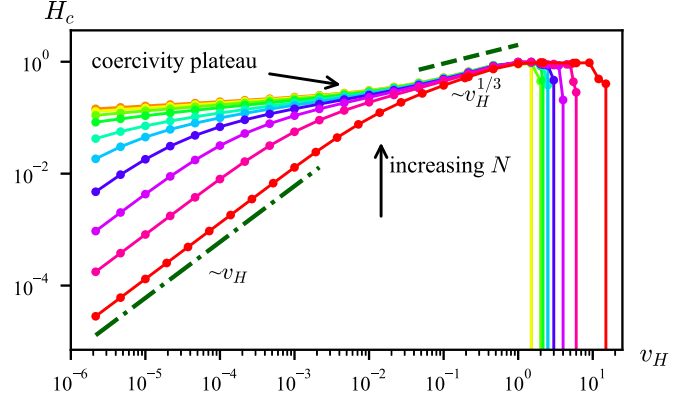


FIG. 8. Coercivity H_c as a function of driving rate v_H for system sizes at $N = 5, 10, 15, 20, 25, 30, 40, 50, 60, 70$ on a log-log scale. Other parameters as in Fig. 7(c).

B. Finite-time scalings

To further characterize the dynamic behavior observed in the hysteresis loops and their dependence on system size and driving rate, we now turn to dissect the finite-time scaling of coercivity in the Curie-Weiss model.

$$I. H_c - H^* \sim v_H^{2/3}$$

During the dynamic process described by Eq. (44) for an infinite-size system, the system evolves along trajectories that may adhere to local equilibrium within metastable states until a sudden transition to a distinct stable state occurs. This transition, contingent on the system's history, exemplifies a field-driven first-order phase transition, driven by the collapse of the metastable state. The spinodal points ($\pm H^*$, $\pm m^*$) are solved from $g(m, H) = 0$ and $\partial_m g(m, H) = 0$ as

$$m^* = -\sqrt{1 - \frac{1}{\beta J}},$$

$$H^* = \frac{1}{\beta} \left[\sqrt{\beta J(\beta J - 1)} - \operatorname{arctanh} \left(\sqrt{\frac{\beta J - 1}{\beta J}} \right) \right]. \quad (45)$$

Notably, when the sudden change occurs at a finite rate, the system's dynamics lags behind the quasistatic evolution, giving rise to an excess work between dynamic and quasistatic hysteresis loops. Such an additional energy cost is obtained in Ref. [15] as

$$w_{\text{ex}} \approx \frac{(-A_1)(1 + \sqrt{1 - \frac{1}{\beta J}})}{[4\sqrt{\beta J(\beta J - 1)} \frac{\beta}{\tau_0}]^{1/3}} v_H^{2/3}, \quad (46)$$

with the first zero point of the Airy function $A_1 \approx -2.338$ [3], which extends to various systems undergoing first-order phase transitions, including open quantum systems [41–43] and chemical reaction networks [44–46]. Although the results above are derived by neglecting fluctuations for infinitely large systems, the authors in Ref. [15] claimed that for large systems with effective nonergodicity, the averaged excess work still scales as $w_{\text{ex}} \sim v_H^{2/3}$, with the quasistatic work still being approximated by that in the deterministic limit. We

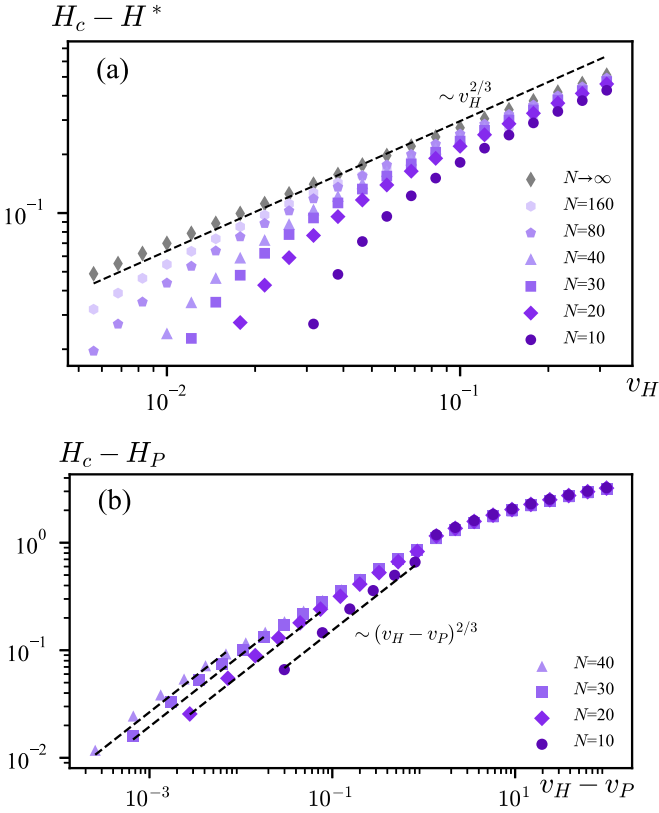


FIG. 9. Demonstrating the 2/3 scaling of increasing coercivity as a function of driving rate v_H in the Curie-Weiss model. Markers in purple are obtained by numerically solving the master equation Eq. (39) in finite-size systems. Gray rhombuses are obtained by numerically solving the deterministic equation Eq. (44) for infinite system. (a) Predictions by the deterministic dynamics. (b) Finite-size corrections for small systems.

further note that, for systems exhibiting FOPT, the excess work can be approximated by the enclosed rectangular area bounded by the upper and lower saturated magnetization m and the interval between H_c and H^* ,

$$w_{\text{ex}} \approx \Delta m (H_c - H^*), \quad (47)$$

where the magnetization difference $\Delta m \approx 2$ across the transition is not related to v_H . This leads to the scaling relation

$$H_c - H^* \sim v_H^{2/3}. \quad (48)$$

Equation (48) is numerically verified in the Fig. 9(a). As shown in this figure, for finite-size systems, the 2/3 scaling of the increasing coercivity is only locally observed: the smaller the system, the more this locality shifts to higher v_H , and the more pronounced the deviation from this scaling becomes. This behavior is closely related to the finite-size effects on the scaling of irreversible work discussed in Ref. [15], which finds that as the system size decreases, the scaling of the average excess work transitions from $v_H^{2/3}$ to v_H . This scaling transition stems from the recovery of ergodicity, where sufficiently strong thermal fluctuations can overcome the energy barrier between the two potential minima. Both observations in our work and in Ref. [15] indicate that finite-size effects fundamentally alter both the scaling behavior of the average

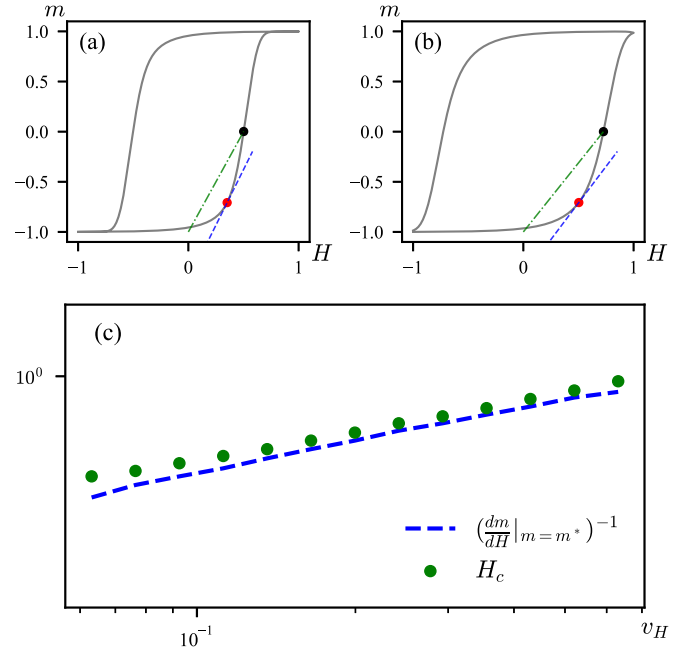


FIG. 10. (a), (b) Numerical observations of two slopes: one at the point $(H^* + v_H \hat{t}_{\text{del}}, m^*)$, and another linking the coercivity to $(0, -1)$. (c) Verifying the closeness of the two slopes. Here, we also choose $N = 30, H_m = 1$.

excess work and coercivity, as well as the dynamic behavior of energy dissipation under external driving. For small systems, a finite-size correction to Eq. (48) is necessary, as we illustrate in Fig. 9(b). A detailed discussion on this issue is provided in Sec. IV C.

2. $H_c \sim v_H^{1/3}$

While the potential landscape transitions from bistable to monostable at $H = H^*$ when $t = t^*$, finite-rate driving induces dynamical lag: the system delays switching to the global minimum by $\hat{t}_{\text{del}}$, defined by $m(t^* + \hat{t}_{\text{del}}) = m^*$. To describe this behavior, one expands the deterministic dynamical equation Eq. (44) around the spinodal point (m^*, H^*) , leading to the approximate evolution equation [15]

$$\frac{d\hat{m}}{d\hat{t}} = \frac{2\beta J \sqrt{\beta J - 1}}{\tau_0} \hat{m}^2 + \frac{2}{\tau_0} \sqrt{\frac{\beta}{J}} v_H \hat{t}, \quad (49)$$

where $\hat{m} = m - m^*$ and $\hat{t} = t - t^*$ are the shifted variables, and the delay time follows as [15]

$$\hat{t}_{\text{del}} = (-A'_1) \left[4\sqrt{\beta J (\beta J - 1)} \frac{\beta}{\tau_0^2} v_H \right]^{-1/3}, \quad (50)$$

with the first zero point of the Airy prime function $A'_1 = -1.019$ (see Appendix C for details).

To further reveal the coercivity scaling connected to Eq. (50), we now turn to investigate the loop slope at the delayed switching point. Numerical observation suggests that the slope at the point $(H^* + v_H \hat{t}_{\text{del}}, m^*)$ corresponding to the delay time $\hat{t}_{\text{del}}$ is approximately equal to the slope connecting the coercivity and the point $(0, -1)$, as shown in Figs. 10(a)

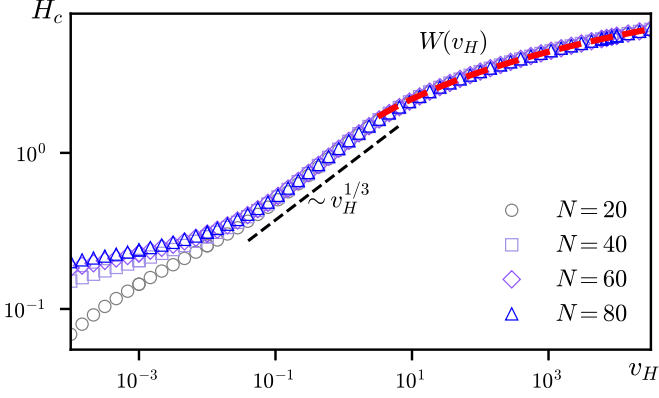


FIG. 11. Detailed scalings in the relatively fast-driving regimes after the plateau within the Curie-Weiss model, where $W(v_H)$ denotes the Lambert W function. In this plot, the driving amplitude is unconstrained to ensure full saturation of the order parameter.

and 10(b), namely,

$$\left. \frac{dm}{dH} \right|_{m=m^*} = v_H^{-1} \left. \frac{d\hat{m}}{d\hat{t}} \right|_{\hat{t}=\hat{t}_{\text{del}}} = \frac{1}{H_c}. \quad (51)$$

Further numerical verification supports this result, as illustrated in Fig. 10(c). By substituting the Airy-function solution Eq. (50) into Eq. (49), the local slope at $\hat{m} = 0$ is obtained as

$$\left. \frac{d\hat{m}}{d\hat{t}} \right|_{\hat{t}=\hat{t}_{\text{del}}} = \frac{2(-A'_1)}{\tau_0} \sqrt{\frac{\beta}{J}} \left[4\sqrt{\beta J(\beta J - 1)} \frac{\beta}{\tau_0^2} \right]^{-1/3} v_H^{2/3}. \quad (52)$$

Rearranging the above equations results in the scaling relation

$$H_c \approx \frac{\tau_0}{2(-A'_1)\sqrt{\beta/J}} \left[4\sqrt{\beta J(\beta J - 1)} \frac{\beta}{\tau_0^2} \right]^{1/3} v_H^{1/3}, \quad (53)$$

which emerges between the coercivity plateau and the fast-driving regime. This $H_c \sim v_H^{1/3}$ scaling is marked with the dashed line in Fig. 11, which is universal for different spin numbers. Interestingly, in the faster driving regime, we notice that H_c deviates from the $1/3$ scaling and still exhibit a universal behavior independent of N . Next, we analyze the behavior of H_c in this rapidly driven regime.

3. $H_c \sim W(v_H)$

In the fast-driving regime, the system's dynamics is dominated by the forced relaxation to monostable state under strong field, considering that the field increases so fast that the system seems to be frozen before the free energy landscape becomes monostable. In the strong-field limit where $|H| \gg |Jm|$, the interaction term Jm becomes negligible compared to H , and the deterministic equation (44) simplifies to

$$\frac{dm}{dt} \approx \frac{2}{\tau_0} [\sinh(\beta H) - m \cosh(\beta H)]. \quad (54)$$

This linear differential equation describes exponential relaxation toward the equilibrium value

$$m_{\text{eq}} \approx \tanh(\beta H) \approx 1 - 2e^{-2\beta H} \quad \text{for } \beta H \gg 1. \quad (55)$$

To estimate the characteristic relaxation time toward equilibrium, we linearize the dynamical equation around m_{eq} :

$$\frac{dm}{dt} \approx -\frac{2}{\tau_0} \cosh(\beta H) (m - m_{\text{eq}}). \quad (56)$$

This leads to the relaxation time

$$\tau_{\text{relax}} \approx \left(\frac{2}{\tau_0} \cosh(\beta H) \right)^{-1} \approx \tau_0 e^{-\beta H}, \quad \text{for } \beta H \gg 1. \quad (57)$$

Unlike comparing the timescales of relaxation and the movement of equilibrium state in the stochastic ϕ^4 model for analyzing the fast-driving regime, in the Curie-Weiss model, the dominant timescale is only limited by τ_{relax} since the equilibrium state infinitely approaches a fixed saturation value in this model. Let τ_c denote the characteristic time at which the system reaches $m = 0$ during the forced relaxation process to the equilibrium state. Considering the relation between the relaxation time and the field strength, we estimate this time by

$$\tau_c \sim e^{-\beta H_c}, \quad (58)$$

Since the external field changes linearly with time, τ_c can also be estimated by

$$\tau_c \sim \frac{H_c}{v_H}. \quad (59)$$

Combining the above two expressions gives the scaling relation:

$$H_c e^{\beta H_c} \sim v_H, \quad (60)$$

with the Lambert W function being the inverse function of xe^x . This scaling behavior is demonstrated by the thick dashed line in Fig. 11.

The different scaling relations in the fast-driving regime between the ϕ^4 and Curie-Weiss models due to fundamental differences in their full dynamical equations, highlight the sensitivity of far-from-equilibrium dynamics to model-specific details, even when the underlying physical picture remains similar. Consequently, different microscopic models may reveal more varied scaling relations in this regime [8,47]. Furthermore, while the Curie-Weiss model exhibits a sustained $1/3$ scaling that bridges the coercivity plateau regime and forced relaxation dynamics at high driving rates, this scaling is largely absent in the stochastic ϕ^4 model, though briefly passed through, which is again attributable to its specific system dynamics.

C. Finite-size scaling of the coercivity plateau

In this subsection, we compare the roles of σ and N in the Fokker-Planck equations of the stochastic ϕ^4 and Curie-Weiss models to bridge the gap between stochastic fluctuations and finite-size effects.

We first derive the Fokker-Planck equation corresponding to the master equation (39) for $N \gg 1$. Substituting the

transition rates (40) into Eq. (39), one obtains

$$\begin{aligned} \frac{\partial P(m, t)}{\partial t} = & -\frac{N}{2\tau_0} [1 - m] e^{\beta[J(m+\frac{1}{N})+H]} P(m, t) \\ & -\frac{N}{2\tau_0} [1 + m] e^{-\beta[J(m-\frac{1}{N})+H]} P(m, t) \\ & +\frac{N}{2\tau_0} \left[1 - \left(m - \frac{2}{N} \right) \right] e^{\beta[J(m-\frac{1}{N})+H]} \\ & \times P\left(m - \frac{2}{N}, t\right) \\ & +\frac{N}{2\tau_0} \left[1 + \left(m + \frac{2}{N} \right) \right] e^{-\beta[J(m+\frac{1}{N})+H]} \\ & \times P\left(m + \frac{2}{N}, t\right). \end{aligned} \quad (61)$$

Using the shift operator [48], the above equation becomes

$$\begin{aligned} \frac{\partial P(m, t)}{\partial t} = & \frac{N}{2\tau_0} \left(e^{-\frac{2}{N} \frac{\partial}{\partial m}} - 1 \right) (1 - m) e^{\beta[Jm+J/N+H]} P(m, t) \\ & + \frac{N}{2\tau_0} \left(e^{\frac{2}{N} \frac{\partial}{\partial m}} - 1 \right) (1 + m) e^{-\beta[Jm-J/N+H]} P(m, t). \end{aligned} \quad (62)$$

Considering $N \gg 1$, and retaining the $1/N$ term in the above equation, one obtains

$$\begin{aligned} \frac{\partial P(m, t)}{\partial t} = & -\frac{2 + 2\beta J/N}{\tau_0} \frac{\partial}{\partial m} [T(m, H) P(m, t)] \\ & + \frac{1}{N} \frac{2}{\tau_0} \frac{\partial^2}{\partial m^2} [S(m, H) P(m, t)], \end{aligned} \quad (63)$$

where $T(m, H) = \sinh \beta(Jm + H) - m \cosh \beta(Jm + H)$ and $S(m, H) = \cosh \beta(Jm + H) - m \sinh \beta(Jm + H)$. Equation (63) is associated with the deterministic equation (44) in the $N \rightarrow \infty$ limit. Expanding Eq. (63) around the spinodal point (m^*, H^*) given in Eq. (45) and keeping the leading order of each term, we obtain

$$\begin{aligned} \frac{\partial P(\hat{m}, t)}{\partial t} \approx & -\frac{2}{\tau_0} \frac{\partial}{\partial \hat{m}} \left[\beta J \sqrt{\beta J - 1} \hat{m}^2 + \sqrt{\frac{\beta}{J}} \hat{H} \right] P(\hat{m}, t) \\ & + \frac{1}{N} \frac{2}{\tau_0} \frac{1}{\sqrt{\beta J}} \frac{\partial^2}{\partial \hat{m}^2} P(\hat{m}, t), \end{aligned} \quad (64)$$

where $\hat{m} \equiv m - m^*$ and $\hat{H} \equiv H - H^*$.

To find the corresponding relation between the system size N in the Curie-Weiss model and the noise strength σ in the stochastic ϕ^4 model, we write down the Fokker-Planck equation for the ϕ^4 model near its spinodal point,

$$\frac{\partial P(\varphi, t)}{\partial t} = -\lambda \frac{\partial}{\partial \varphi} [(-a_3 \varphi^2 + h) P(\varphi, t)] + \lambda \sigma^2 \frac{\partial^2}{\partial \varphi^2} P(\varphi, t), \quad (65)$$

with φ , h , and a_3 defined in Eq. (23). Comparing the coefficient of the second diffusion part in the above equation and that in Eq. (64), one obtains $\sigma^2 \sim N^{-1}$. It is worth noting that this relation is independent of the specific transition rates used in the Curie-Weiss model [5,28] and is also observed in different systems [30–32]. Therefore, it follows from Eq. (29)

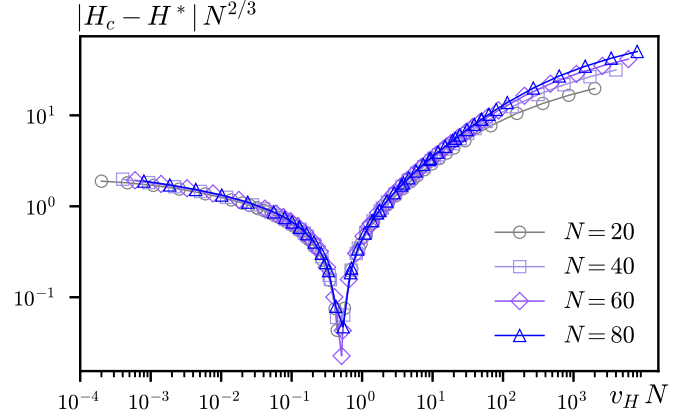


FIG. 12. Finite-size scaling collapse of coercivity plateau with saturated hysteresis loops.

that

$$H^* - H_p \sim N^{-2/3} \quad \text{at} \quad v_p \sim 1/N. \quad (66)$$

According to Eq. (66), one can rescale the axes in the $H_c - v_H$ plot. As shown in Fig. 12, the coercivity curves for different system sizes ($N = 20, 40, 60, 80$) approximately collapse onto a single universal curve, confirming the validity of the finite-size scaling relation. Notably, as the finite-size scaling analysis is performed near the coercivity plateau, minor deviations arise among curves of different N at both ends far from the plateau. It is indicated from the figure that the universal curve diverges at a critical driving rate that yields $H_c = H^*$. In the infinite-size limit, this critical driving rate approaches zero, rendering the region to its left inaccessible. This demonstrates that the coercivity plateau, which does not vanish in the slow-driving regime, results from taking the thermodynamic limit ($N \rightarrow \infty$) first, namely, $\lim_{v_H \rightarrow 0} \lim_{N \rightarrow \infty} H_c = H^*$. In contrast, for finite-size systems, coercivity values span both sides of this critical point, with the gently varying region at lower driving rates reflecting the presence of a coercivity plateau.

Due to the deviation of the coercivity plateau from the spinodal field H^* in finite-size systems, a finite-size correction to the $2/3$ scaling [Eq. (48)] is obtained by associating Eqs. (37) and (66). Using an operational definition of the plateau point—the minimum point of $\partial \ln(H_c)/\partial \ln(v_H)$ —this corrected postplateau scaling is illustrated in Fig. 9(b), and we note that it is more robust than the uncorrected one [Fig. 9(a)] for small systems.

V. CONCLUSION AND OUTLOOK

In this work, we have developed a unified theoretical framework to characterize coercivity in dynamic hysteresis associated with FOPT under periodic driving, spanning the full range from near-equilibrium to fast-driving regimes, and covering both of deterministic and stochastic dynamics. Building on the coercivity landscape proposed in our companion paper [23] for the stochastic ϕ^4 model, we provided analytical derivations and numerical demonstrations, focusing on scaling behaviors with respect to driving rates as well as noise strengths through detailed calculations. By

systematically studying the behavior of coercivity H_c across noise strengths σ , we found a nonmonotonic trend in its scaling exponent with respect to the driving rate v_H : the exponent first decreases and then increases, confirming the presence of a coercivity plateau (Fig. 3). We analyzed in depth two limiting scaling regimes: $H_c \sim v_H$ in the near-equilibrium regime, and $H_c \sim v_H^{1/2}$ in the fast-driving regime. To further characterize the plateau related to effective dynamical ergodicity breaking, we explored its scaling behavior with respect to σ via renormalization-group theory [Eq. (29)] and analyzed the postplateau finite-time scaling [Eq. (37)], with numerical validation provided in [23].

We further applied this framework to the Curie-Weiss model, solving its master equations to study hysteresis loop morphologies under various conditions, including spin number N , driving amplitude H_m , and driving rate v_H . Given the analogous roles of noise strength σ and system size N in mean-field models, we particularly analyzed the role of finite system size by constructing the full coercivity landscape for various N . Comparing the coercivity landscape of the stochastic ϕ^4 and Curie-Weiss models, we identified several universal features: (i) the existence of the coercivity plateau (Fig. 8); (ii) the linear finite-time scaling $H_c \sim v_H$ before the coercivity plateau; (iii) the $2/3$ -exponent finite-time scaling just beyond the plateau (Fig. 9); (iv) the finite-size scaling of the plateau with a $\sigma^{-2} \sim N$ correspondence, which is derived by comparing the Fokker-Planck equations of both models. However, in the fast-driving regime, despite similar mechanism, the Curie-Weiss model shows different scaling behavior, following the Lambert W-function, due to differences in the underlying dynamics. Notably, an intermediate scaling $H_c \sim v_H^{1/3}$ emerges in the Curie-Weiss model between the plateau and fast-driving regimes, which is only briefly traversed in the ϕ^4 model and does not persist over observable timescales. These findings highlight the sensitivity of far-from-equilibrium dynamics to model-specific details.

The insights presented here help clarify the ambiguous connections between experimental observations and theoretical predictions (see the end matter of the companion paper [23]). Altogether, this work lays a foundation for a unified understanding of dynamic hysteresis and motivates future research aimed at connecting phenomenological models with fundamental statistical physics frameworks, with specific remarks available in the conclusion of the companion paper [23]. In addition to arising from finite-size effects in the Curie-Weiss model, the noise scaling in the mesoscopic ϕ^4 model may also describe fluctuations in systems with short-range interactions, especially those relevant to nucleation processes over potential barriers [6,17,49,50]. Finally, exploring dynamical differences away from the coercivity plateau between the stochastic ϕ^4 and Curie-Weiss models is of interest, as it may further clarify how the free-energy landscape influences dynamical scaling behaviors. Two open questions naturally arise: i) How does the choice of transition rate in the Curie-Weiss model affect scaling in the fast-driving regime? Our results unveil that the fast-driving scaling exponent is model dependent. Since transition rates satisfying detailed balance are not unique [28], different choices may yield distinct fast-driving scalings even for the same free-energy functional;

ii) What is the complete scaling behavior under covarying parameters in the Curie-Weiss model? Although we have demonstrated consistency between fixed-parameter scaling and complete scaling in the stochastic ϕ^4 model, identifying the corresponding complete scaling in the Curie-Weiss case would complement the multiple scaling regimes shown in Fig. 8. This behavior may differ due to the fixed saturation of the order parameter in the Curie-Weiss model, in contrast to the parameter- and field-dependent saturation in the ϕ^4 model, where higher-order terms of Eq. (44) beyond the lowest-order expansion in Eq. (49) are expected to play an important role.

ACKNOWLEDGMENTS

We thank Sai Li for the helpful discussions in the early stage of this work. This work is supported by the National Natural Science Foundation of China under Grant No. 12305037 and the Fundamental Research Funds for the Central Universities under Grant No. 2023NTST017.

DATA AVAILABILITY

The data that support the findings of this article are not publicly available upon publication because it is not technically feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

APPENDIX A: UNIVERSALITY OF THE $1/2$ SCALING IN THE STOCHASTIC ϕ^4 MODEL

The $1/2$ scaling in the fast-driving regime is not merely occasional but universal for different values of σ , a_2 , and a_4 . While its independence on σ is demonstrated in Fig. 3(a), we further illustrate its independence on a_2 and a_4 by showing coercivity curves obtained from numerically solving Eq. (4) in the deterministic limit ($\sigma = 0$), as depicted in Fig. 13. In this plot, for simplicity, the coercivity is extracted from monotonically increasing-field driving rather than periodic driving. The initial field value is chosen slightly beyond the spinodal point at the given a_2 and a_4 , and the system is initialized in the corresponding equilibrium state. The rate range for observing $1/2$ scaling is more sensitive to variations in a_2 than in a_4 . This is because a_2 's influence on the strong field assumption $|H| \gg \sqrt{|a_2^3/a_4|}$ scales with $a_2^{3/2}$ (Sec. III C), significantly amplifying its impact compared to a_4 's inverse square root dependence.

APPENDIX B: THE CONSISTENCY BETWEEN THE CURRENT WORK AND REF. [40]

In Ref. [40], Zhong proposed a systematic framework for establishing complete scaling relations for finite-time first-order phase transitions, including the stochastic ϕ^4 model, where the coercivity scaling exponent is $2/3$. We demonstrate here that the specific scaling exponent of coercivity with respect to the driving rate is dependent on the values of parameters σ , a_2 , and a_4 . The linear scaling in the

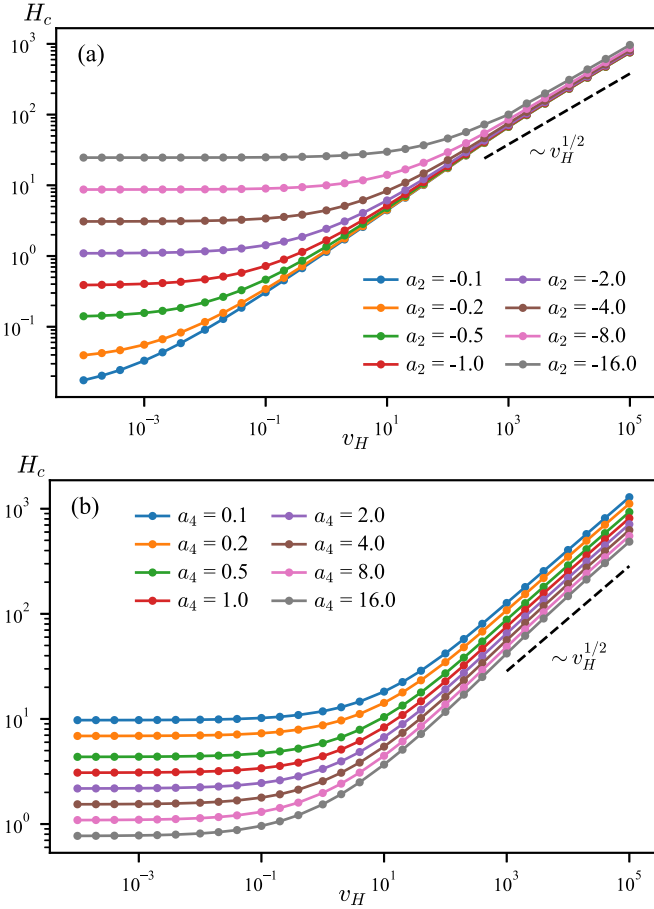


FIG. 13. Universality of the 1/2 scaling over broad a_2 and a_4 . $a_4 = 1$ and $a_2 = -4$ are respectively fixed in panels (a), (b). In this plot, $\lambda = 1$.

near-equilibrium regime and 1/2 scaling in the fast-driving regime presented in the current work do not contradict the complete universal 2/3 scaling given in Ref. [40]; rather, they emerge under different parametric conditions. First, for fixed σ , a_2 , and a_4 , the linear and 1/2 scaling are explicitly demonstrated in Fig. 3(a) and analyzed in the main text. Second, we respectively select two reference points located in the linear and 1/2-scaling regimes in Fig. 3(a) and then covary σ , a_2 , and a_4 according to the complete scaling exponents provided in Ref. [40]. The resulting hysteresis curves are displayed in Fig. 14: panels (a) and (c) show the original hysteresis loops under covarying parameters (σ , a_2 , a_4 , and v_H), while panels (b) and (d) present the corresponding curves in the rescaled H - ϕ coordinate system. Under such covarying conditions, the emerged coercivity scaling is exactly 2/3, aligning with Ref. [40].

APPENDIX C: DERIVATION OF THE DELAY TIME OF STATE TRANSITIONS

To isolate the universal structure of the dynamic equation Eq. (49), we introduce dimensionless variables $u = \alpha \hat{m}$, $s = \gamma \hat{t}$, with the two scale factors α , $\gamma > 0$ chosen to

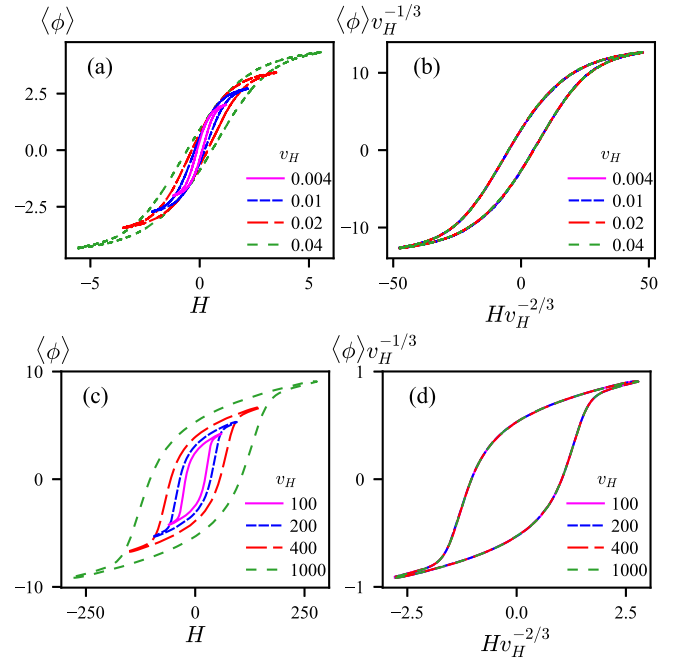


FIG. 14. Collapse of hysteresis loops under the complete scaling according to the covariant parameter set in Ref. [40]: $\sigma = \sigma^{\text{ref}} (v_H/v_H^{\text{ref}})^{1/2}$, $a_2 = a_2^{\text{ref}} (v_H/v_H^{\text{ref}})^{1/3}$, and $a_4 = a_4^{\text{ref}} (v_H/v_H^{\text{ref}})^{-1/3}$. (a), (b) The reference point in the parameter space is $(\sigma^{\text{ref}} = 1.0, a_2^{\text{ref}} = -4, a_4^{\text{ref}} = 1, v_H^{\text{ref}} = 0.004)$, which is in the near-equilibrium regime in Fig. 3(a). The loops in panel (a) collapse to a unique one in panel (b). (c), (d) The reference point in the parameter space is $(\sigma^{\text{ref}} = 1.0, a_2^{\text{ref}} = -4, a_4^{\text{ref}} = 1, v_H^{\text{ref}} = 100)$, which is in the 1/2-scaling regime in Fig. 3(a). The loops in panel (c) collapse to a unique one in panel (d). In this plot, $\lambda = 1$.

eliminate all prefactors in the resulting equation. In this case, one has

$$\frac{du}{ds} = \frac{2\beta J \sqrt{\beta J - 1}}{\tau_0 \alpha \gamma} u^2 + \frac{2\alpha}{\tau_0 \gamma^2} \sqrt{\frac{\beta}{J}} v_H s. \quad (\text{C1})$$

To rescale the coefficients to unity, we have

$$\frac{2\beta J \sqrt{\beta J - 1}}{\tau_0 \alpha \gamma} = 1, \quad \frac{2\alpha v_H}{\tau_0 \gamma^2} \sqrt{\frac{\beta}{J}} = 1, \quad (\text{C2})$$

which leads to the following expressions:

$$\alpha = \left(\frac{2}{\tau_0} \beta J \sqrt{\beta J - 1} \right)^{2/3} \left(\frac{2}{\tau_0} \sqrt{\frac{\beta}{J}} v_H \right)^{-1/3}, \quad (\text{C3})$$

$$\gamma = \left(\frac{2}{\tau_0} \beta J \sqrt{\beta J - 1} \right)^{1/3} \left(\frac{2}{\tau_0} \sqrt{\frac{\beta}{J}} v_H \right)^{1/3}.$$

With these rescaled coefficients, the original equation simplifies to the standard Riccati form $du/ds = u^2 + s$, the solution to which, under the asymptotic boundary condition $s \rightarrow -\infty$ and $u \rightarrow \sqrt{-s}$, is given by the Airy function as $u(s) = \text{Ai}'(-s)/\text{Ai}(-s)$ [3]. Thus, the delay time is obtained as Eq. (50) in the main text.

- [1] B. K. Chakrabarti and M. Acharyya, Dynamic transitions and hysteresis, *Rev. Mod. Phys.* **71**, 847 (1999).
- [2] K. A. Morris, What is hysteresis? *Appl. Mech. Rev.* **64**, 050801 (2011).
- [3] P. Jung, G. Gray, R. Roy, and P. Mandel, Scaling law for dynamical hysteresis, *Phys. Rev. Lett.* **65**, 1873 (1990).
- [4] D. C. Brody and D. W. Hook, Information geometry in vapour-liquid equilibrium, *J. Phys. A: Math. Theor.* **42**, 023001 (2009).
- [5] T. Mori, S. Miyashita, and P. A. Rikvold, Asymptotic forms and scaling properties of the relaxation time near threshold points in spinodal-type dynamical phase transitions, *Phys. Rev. E* **81**, 011135 (2010).
- [6] K. Binder, Theory of first-order phase transitions, *Rep. Prog. Phys.* **50**, 783 (1987).
- [7] M. M. Campagna, J. Hrubý, M. E. H. van Dongen, and D. M. J. Smeulders, Homogeneous water nucleation: Experimental study on pressure and carrier gas effects, *J. Chem. Phys.* **153**, 164303 (2020).
- [8] M. Rao, H. R. Krishnamurthy, and R. Pandit, Magnetic hysteresis in two model spin systems, *Phys. Rev. B* **42**, 856 (1990).
- [9] Q. Jiang, H.-N. Yang, and G.-C. Wang, Scaling and dynamics of low-frequency hysteresis loops in ultrathin Co films on a Cu(001) surface, *Phys. Rev. B* **52**, 14911 (1995).
- [10] S. W. Sides, P. A. Rikvold, and M. A. Novotny, Kinetic Ising model in an oscillating field: Avrami theory for the hysteretic response and finite-size scaling for the dynamic phase transition, *Phys. Rev. E* **59**, 2710 (1999).
- [11] T. A. Moore and J. A. C. Bland, Mesofrequency dynamic hysteresis in thin ferromagnetic films, *J. Phys.: Condens. Matter* **16**, R1369 (2004).
- [12] Z. Geng, K. J. H. Peters, A. A. P. Trichet, K. Malmir, R. Kolkowski, J. M. Smith, and S. R. K. Rodriguez, Universal scaling in the dynamic hysteresis, and non-Markovian dynamics, of a tunable optical cavity, *Phys. Rev. Lett.* **124**, 153603 (2020).
- [13] G. H. Goldsztein, F. Broner, and S. H. Strogatz, Dynamical hysteresis without static hysteresis: Scaling laws and asymptotic expansions, *SIAM J. Appl. Math.* **57**, 1163 (1997).
- [14] C. N. Luse and A. Zangwill, Discontinuous scaling of hysteresis losses, *Phys. Rev. E* **50**, 224 (1994).
- [15] Y.-X. Wu, J.-F. Chen, and H. T. Quan, Ergodicity breaking and scaling relations for finite-time first-order phase transition, *Phys. Rev. Lett.* **134**, 177101 (2025).
- [16] A. Banerjee and T. Bar, Finite-dimensional signature of spinodal instability in an athermal hysteretic transition, *Phys. Rev. B* **107**, 024103 (2023).
- [17] S. Kundu, R. K. Patel, S. Middey, and B. Bansal, Dynamic hysteresis at a noisy saddle node shows power-law scaling but nonuniversal exponent, *Phys. Rev. E* **108**, 024101 (2023).
- [18] M. C. Mahato and S. R. Shenoy, Hysteresis as rate competition: A Landau model example, *Physica A* **186**, 220 (1992).
- [19] P. A. Rikvold, H. Tomita, S. Miyashita, and S. W. Sides, Metastable lifetimes in a kinetic Ising model: Dependence on field and system size, *Phys. Rev. E* **49**, 5080 (1994).
- [20] T. Tomé and M. J. de Oliveira, Dynamic phase transition in the kinetic Ising model under a time-dependent oscillating field, *Phys. Rev. A* **41**, 4251 (1990).
- [21] D. T. Robb, Y. H. Xu, O. Hellwig, J. McCord, A. Berger, M. A. Novotny, and P. A. Rikvold, Evidence for a dynamic phase transition in [Co/Pt]₃ magnetic multilayers, *Phys. Rev. B* **78**, 134422 (2008).
- [22] P. Riego, P. Vavassori, and A. Berger, Towards an understanding of dynamic phase transitions, *Physica B* **549**, 13 (2018).
- [23] M. Chen, X.-H. Zhao, and Y.-H. Ma, companion paper, Coercivity landscape characterizes dynamic hysteresis, *Phys. Rev. Lett.* **136**, 117102 (2026).
- [24] J. R. Tucker and B. I. Halperin, Onset of superconductivity in one-dimensional systems, *Phys. Rev. B* **3**, 3768 (1971).
- [25] S. Gong, F. Zhong, X. Huang, and S. Fan, Finite-time scaling via linear driving, *New J. Phys.* **12**, 043036 (2010).
- [26] S. Friedli and Y. Velenik, *Statistical Mechanics of Lattice Systems: A Concrete Mathematical Introduction*, 1st ed. (Cambridge University Press, Cambridge, New York, 2017).
- [27] J. Meibohm and M. Esposito, Finite-time dynamical phase transition in nonequilibrium relaxation, *Phys. Rev. Lett.* **128**, 110603 (2022).
- [28] E. R. Fiori, C. Maes, and R. Vids, Specific heat of the driven Curie-Weiss model, *J. Stat. Phys.* **192**, 63 (2025).
- [29] N. Berglund and B. Gentz, *Noise-Induced Phenomena in Slow-Fast Dynamical Systems: A Sample-Paths Approach* (Springer, London, 2006).
- [30] Z. Chen and C. C. Yu, Measurement-noise maximum as a signature of a phase transition, *Phys. Rev. Lett.* **98**, 057204 (2007).
- [31] H. S. Leff, Fluctuations in particle number for a photon gas, *Am. J. Phys.* **83**, 362 (2015).
- [32] Z. Fei and Y.-H. Ma, Temperature fluctuations in mesoscopic systems, *Phys. Rev. E* **109**, 044101 (2024).
- [33] L. E. Reichl, *A Modern Course in Statistical Physics*, 4th ed. (Wiley-VCH, Weinheim, Germany, 2016).
- [34] A. C. Brańka and D. M. Heyes, Algorithms for Brownian dynamics simulation, *Phys. Rev. E* **58**, 2611 (1998).
- [35] H. Risken, *The Fokker-Planck Equation: Methods of Solution and Applications* (Springer, Berlin, Heidelberg, 1996).
- [36] S. Chikazumi, Mechanism of high coercivity in rare-earth permanent magnets, *J. Magn. Magn. Mater.* **54-57**, 1551 (1986).
- [37] X. Pei, A. C. Smith, L. Vandenbossche, and J. Rens, Magnetic characterization of soft magnetic cores at cryogenic temperatures, *IEEE Trans. Appl. Supercond.* **29**, 1 (2019).
- [38] K. K. M. Pandey, J. S. Chen, G. M. Chow, and J. F. Hu, L₁CoPtTa₂O₅ exchange coupled multilayer media for magnetic recording, *Appl. Phys. Lett.* **94**, 232502 (2009).
- [39] I. Tagawa and Y. Nakamura, Relationships between high density recording performance and particle coercivity distribution, *IEEE Trans. Magn.* **27**, 4975 (1991).
- [40] F. Zhong, Complete universal scaling in first-order phase transitions, *Chin. Phys. Lett.* **41**, 100502 (2024).
- [41] P. Drummond and D. Walls, Quantum theory of optical bistability. I. Nonlinear polarisability model, *J. Phys. A: Math. Gen.* **13**, 725 (1980).
- [42] A. Alodjants, P. Zacharenko, and D. Tsarev, Superradiant phase transitions in ultrastrong coupling regime, *Laser Phys. Lett.* **21**, 125205 (2024).
- [43] Xin H. H. Zhang and H. U. Baranger, Driven-dissipative phase transition in a Kerr oscillator: From semiclassical PT symmetry to quantum fluctuations, *Phys. Rev. A* **103**, 033711 (2021).
- [44] B. Nguyen and U. Seifert, Exponential volume dependence of entropy-current fluctuations at first-order phase transitions in chemical reaction networks, *Phys. Rev. E* **102**, 022101 (2020).
- [45] F. Schlögl, Chemical reaction models for non-equilibrium phase transitions, *Z. Phys.* **253**, 147 (1972).

- [46] H. Ge and H. Qian, Non-equilibrium phase transition in mesoscopic biochemical systems: From stochastic to non-linear dynamics and beyond, *J. R. Soc. Interface* **8**, 107 (2011).
- [47] P. Shukla, Hysteresis in the Ising model with Glauber dynamics, *Phys. Rev. E* **97**, 062127 (2018).
- [48] W. Paul, D. W. Heermann, and K. Binder, Relaxation of metastable states in finite mean-field kinetic Ising systems, *J. Phys. A: Math. Gen.* **22**, 3325 (1989).
- [49] A. Pelissetto and E. Vicari, Dynamic off-equilibrium transition in systems slowly driven across thermal first-order phase transitions, *Phys. Rev. Lett.* **118**, 030602 (2017).
- [50] F. Zhong, Universal scaling in first-order phase transitions mixed with nucleation and growth, *J. Phys.: Condens. Matter* **30**, 445401 (2018).